



You are a Data Analyst working with the X for Advertisers team to evaluate the effectiveness of advertising campaigns across various audience segments....

Question 1: What is the monthly total engagement count for each audience segment in April 2024 and May 2024? Hint: there is some messy data in the audience segment column, so you might want to clean that up first.

```
SELECT LOWER(audience_segment),  
       TO_CHAR(engagement_date, 'YYYY-MM') AS month_num,  
       SUM(engagement_count) AS total_engagement  
FROM fct_campaign_engagement  
WHERE engagement_date >= '2024-04-01' AND engagement_date < '2024-06-01'  
GROUP BY LOWER(audience_segment), TO_CHAR(engagement_date, 'YYYY-MM')
```

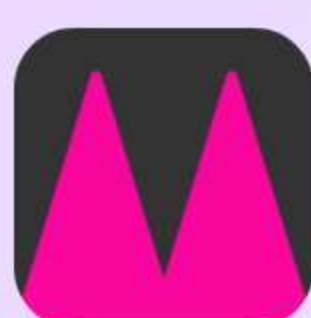




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Question 2: What is the monthly engagement growth rates from April 2024 to May 2024? This calculation will help the X for Advertisers team understand basic engagement performance trends across audience groups for campaign optimization.

```
WITH total AS (  
  SELECT  
    INITCAP(audience_segment) AS audience_segment,  
    TO_CHAR(engagement_date, 'YYYY-MM') AS month,  
    SUM(engagement_count) AS total_engagement  
  FROM fct_campaign_engagement  
  WHERE TO_CHAR(engagement_date, 'YYYY-MM') IN ('2024-04', '2024-05')  
  GROUP BY INITCAP(audience_segment), TO_CHAR(engagement_date, 'YYYY-MM')  
)  
  
with_growth AS (  
  SELECT  
    audience_segment,  
    month,  
    total_engagement,  
    LEAD(total_engagement) OVER (PARTITION BY audience_segment  
ORDER BY month) AS next_month_engagement  
  FROM total  
)  
  
SELECT  
  audience_segment,  
  total_engagement AS april,  
  next_month_engagement AS may,  
  ROUND(  
    CASE  
      WHEN total_engagement = 0 THEN NULL  
      ELSE (next_month_engagement - total_engagement) * 100.0  
    / total_engagement  
    END, 2  
  ) AS growth_rate_percent  
FROM with_growth  
WHERE month = '2024-04'  
ORDER BY 4 DESC
```





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Question 3: Which audience segments had the highest engagement growth rate from April 2024 to May 2024, and which had the lowest? Don't forget to clean up the inconsistent capitalization in the audience segment column.

Identifying these trends will help the X for Advertisers team optimize targeting strategies by focusing on the most responsive audience segments.

```
WITH total AS (
  SELECT
    INITCAP(audience_segment) AS audience_segment,
    TO_CHAR(engagement_date, 'YYYY-MM') AS month,
    SUM(engagement_count) AS total_engagement
  FROM fct_campaign_engagement
  WHERE TO_CHAR(engagement_date, 'YYYY-MM') IN ('2024-04', '2024-05')
  GROUP BY INITCAP(audience_segment), TO_CHAR(engagement_date, 'YYYY-MM')),

with_growth AS (
  SELECT
    audience_segment,
    month,
    total_engagement,
    LEAD(total_engagement) OVER (PARTITION BY audience_segment
ORDER BY month) AS next_month_engagement
  FROM total),

ranking AS(
SELECT
  audience_segment,
  total_engagement AS april,
  next_month_engagement AS may,
  ROUND(
    CASE
      WHEN total_engagement = 0 THEN NULL
      ELSE (next_month_engagement - total_engagement) * 100.0
/ total_engagement
    END, 2
  ) AS growth_rate_percent
FROM with_growth
WHERE month = '2024-04'
ORDER BY 4 DESC),

ranked AS (
  SELECT *,
    RANK() OVER (ORDER BY growth_rate_percent DESC) AS rank_highest,
    RANK() OVER (ORDER BY growth_rate_percent ASC) AS rank_lowest
  FROM ranking)

SELECT *
FROM ranked
WHERE rank_highest = 1 OR rank_lowest = 1;
```

